

v0.1 Search-Space Tightening

The original search initializes the incumbent radius with the first basis vector,

$$R_0 = \|b_1\|.$$

However, LLL does not guarantee that b_1 is the shortest vector already present in the basis. Therefore the initial search radius can be unnecessarily large.

The improved initialization scans the complete basis first:

$$i^* = \operatorname{argmin}_i \|b_i\|, \quad R_0 = \|b_{i^*}\|, \quad z_0 = e_{i^*}.$$

Thus MCTS starts from the best vector already known without spending search nodes rediscovering an existing basis vector.

Immediate Radius Tightening

For a partial coefficient assignment, legal actions satisfy

$$\rho_{k+1} + g_k(z_k - c_k)^2 < R^2.$$

Whenever an exact terminal vector improves the incumbent,

$$\|v\| < R,$$

the global radius is updated immediately:

$$R \leftarrow \|v\|.$$

The legal intervals of all active nodes are then tightened using the new radius. Branches that can no longer satisfy the exact GSO bound are disabled immediately.

Therefore the search loop becomes

exact improvement $\rightarrow$ update $R \rightarrow$ tighten legal space $\rightarrow$ continue MCTS.

Hint: Facing with higher dimensions (such as $\dim > 90$), one SV $\rightarrow$ one R update might be too frequent (or no impact), the global update policy requires to be further derived and verified.